Web-based Supplementary Materials for Halton Iterative Partitioning: spatially balanced sampling via partitioning by

B. L. Robertson[1,*], T. McDonald[2], C. J. Price[1] and J. A. Brown[1]

[1]School of Mathematics and Statistics, University of Canterbury, Private Bag 4800, Christchurch, New Zealand

[2]Western EcoSystems Technology, Inc. Cheyenne, Wyoming 82001, USA

*email: blair.robertson@canterbury.ac.nz

Table 1: The number of boxes, $B = 2^{J_1}3^{J_2}$, that were considered to partition a point resource under HIP sampling.

B	J_1	J_2	B	J_1	J_2	B	J_1	J_2	B	J_1	J_2
6	1	1	144	4	2	1728	6	3	10368	7	4
12	2	1	216	3	3	2592	5	4	20736	8	4
24	3	1	288	5	2	3456	7	3	31104	7	5
36	2	2	432	4	3	5184	6	4	62208	8	5
72	3	2	864	5	3	7776	5	5			

1 Permutation of Halton Indices

The Halton index k (mod $B = 2^{J_1}3^{J_2}$) for a particular Halton box

$$[\ell_1, u_1) \times [\ell_2, u_2), \tag{1}$$

in $[0,1)^2$ is obtained by solving the system of congruences

$$\begin{aligned} k &= a_1 \pmod{2^{J_1}} \\ k &= a_2 \pmod{3^{J_2}}, \end{aligned} \tag{2}$$

where

$$a_i = \sum_{j=1}^{J_i} \left\{ \lfloor \ell_i b_i^j \rfloor \mod b_i \right\} b_i^{j-1}.$$

For example, the mod 36 index for Halton box $[1/4, 1/2) \times [1/3, 4/9)$ is 10 (see the first entry of Table 2 for all the mod 36 indices).

The Halton indices are permuted by changing the $a_i \in \{0, 1, \ldots, b_i^{J_i} - 1\}$ values in (2). First, the values $\{0, 1, \ldots, b_i^{J_i} - 1\}$ are permuted for each base using Algorithm 1 to give $\mathcal{P}_1$ and $\mathcal{P}_2$. The a_i values for box (1) are replaced with

$$a_1 = \mathcal{P}_1^{(2^{J_1}\ell_1+1)} \quad \text{and} \quad a_2 = \mathcal{P}_2^{(3^{J_2}\ell_2+1)},$$

where $\mathcal{P}^{(i)}$ denotes the ith element in $\mathcal{P}$. The system of congruences (2) is then solved to give the permuted Halton index k (mod B) for box (1). Consider, for example, $B = 36$ and the mod B Halton box $[1/4, 1/2) \times [1/3, 4/9)$. If

$$\mathcal{P}_1 = \{0, 2, 1, 3\} \quad \text{and} \quad \mathcal{P}_2 = \{7, 4, 1, 2, 8, 5, 6, 0, 3\}, \tag{3}$$

the box's permuted Halton index k is a solution to

$$\begin{aligned} k &= 2 \pmod{4} \\ k &= 2 \pmod{9}. \end{aligned}$$

Algorithm 1 Permutation of $\{0, 1, \ldots, b_i^{J_i} - 1\}$, where $J_i > 1$. Here σ_b is a row vector containing a random permutation of $\{0, 1, \ldots, b_i - 1\}$ and $\mathcal{I}_j$ is a row vector containing b_i copies of the jth element in $\mathcal{I}$.

$\mathcal{I} = \sigma_b$
for $k = 1$ to $J_i - 1$ **do**
  $v = (\,)$
  for $j = 1$ to b_i^k **do**
    $v \leftarrow \big(v, \mathcal{I}_j + b_i^k \sigma_b\big)$
  end for
  $\mathcal{I} \leftarrow v$
end for
Output $\mathcal{P}_i = \mathcal{I}$.

Hence, the permuted index for this box is $k = 2 \pmod{36}$ (the original index was 10). Two different permutations of this form are given in rows two and three of Table 2, where the first permutation uses (3) and the second uses

$$\mathcal{P}_1 = \{1, 3, 2, 0\} \quad \text{and} \quad \mathcal{P}_2 = \{0, 3, 6, 5, 2, 8, 7, 1, 4\}.$$

Table 2: Halton indices for 36 Halton boxes and their corresponding mod B nested structure, where row i column j is the Halton index for Halton box $[(j-1)/4, j/4) \times [(9-i)/9, (10-i)/9)$. Two permutations of the indices are shown.

	mod 36	mod 12	mod 6	mod 2
Original	$\begin{pmatrix} 8 & 26 & 17 & 35 \\ 32 & 14 & 5 & 23 \\ 20 & 2 & 29 & 11 \\ 16 & 34 & 25 & 7 \\ 4 & 22 & 13 & 31 \\ 28 & 10 & 1 & 19 \\ 24 & 6 & 33 & 15 \\ 12 & 30 & 21 & 3 \\ 0 & 18 & 9 & 27 \end{pmatrix}$	$\begin{pmatrix} 8 & 2 & 5 & 11 \\ 8 & 2 & 5 & 11 \\ 8 & 2 & 5 & 11 \\ 4 & 10 & 1 & 7 \\ 4 & 10 & 1 & 7 \\ 4 & 10 & 1 & 7 \\ 0 & 6 & 9 & 3 \\ 0 & 6 & 9 & 3 \\ 0 & 6 & 9 & 3 \end{pmatrix}$	$\begin{pmatrix} 2 & 2 & 5 & 5 \\ 2 & 2 & 5 & 5 \\ 2 & 2 & 5 & 5 \\ 4 & 4 & 1 & 1 \\ 4 & 4 & 1 & 1 \\ 4 & 4 & 1 & 1 \\ 0 & 0 & 3 & 3 \\ 0 & 0 & 3 & 3 \\ 0 & 0 & 3 & 3 \end{pmatrix}$	$\begin{pmatrix} 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{pmatrix}$
Permutation 1	$\begin{pmatrix} 12 & 30 & 21 & 3 \\ 0 & 18 & 9 & 27 \\ 24 & 6 & 33 & 15 \\ 32 & 14 & 5 & 23 \\ 8 & 26 & 17 & 35 \\ 20 & 2 & 29 & 11 \\ 28 & 10 & 1 & 19 \\ 4 & 22 & 13 & 31 \\ 16 & 34 & 25 & 7 \end{pmatrix}$	$\begin{pmatrix} 0 & 6 & 9 & 3 \\ 0 & 6 & 9 & 3 \\ 0 & 6 & 9 & 3 \\ 8 & 2 & 5 & 11 \\ 8 & 2 & 5 & 11 \\ 8 & 2 & 5 & 11 \\ 4 & 10 & 1 & 7 \\ 4 & 10 & 1 & 7 \\ 4 & 10 & 1 & 7 \end{pmatrix}$	$\begin{pmatrix} 0 & 0 & 3 & 3 \\ 0 & 0 & 3 & 3 \\ 0 & 0 & 3 & 3 \\ 2 & 2 & 5 & 5 \\ 2 & 2 & 5 & 5 \\ 2 & 2 & 5 & 5 \\ 4 & 4 & 1 & 1 \\ 4 & 4 & 1 & 1 \\ 4 & 4 & 1 & 1 \end{pmatrix}$	$\begin{pmatrix} 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{pmatrix}$
Permutation 2	$\begin{pmatrix} 13 & 31 & 22 & 4 \\ 1 & 19 & 10 & 28 \\ 25 & 7 & 34 & 16 \\ 17 & 35 & 26 & 8 \\ 29 & 11 & 2 & 20 \\ 5 & 23 & 14 & 32 \\ 33 & 15 & 6 & 24 \\ 21 & 3 & 30 & 12 \\ 9 & 27 & 18 & 0 \end{pmatrix}$	$\begin{pmatrix} 1 & 7 & 10 & 4 \\ 1 & 7 & 10 & 4 \\ 1 & 7 & 10 & 4 \\ 5 & 11 & 2 & 8 \\ 5 & 11 & 2 & 8 \\ 5 & 11 & 2 & 8 \\ 9 & 3 & 6 & 0 \\ 9 & 3 & 6 & 0 \\ 9 & 3 & 6 & 0 \end{pmatrix}$	$\begin{pmatrix} 1 & 1 & 4 & 4 \\ 1 & 1 & 4 & 4 \\ 1 & 1 & 4 & 4 \\ 5 & 5 & 2 & 2 \\ 5 & 5 & 2 & 2 \\ 5 & 5 & 2 & 2 \\ 3 & 3 & 0 & 0 \\ 3 & 3 & 0 & 0 \\ 3 & 3 & 0 & 0 \end{pmatrix}$	$\begin{pmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \end{pmatrix}$

Table 3: Results for populations 1, 2 and 3 with $N = 10,000$ for different sample sizes, n. The reported values are averages using 1000 different samples, where $\mathrm{V}_{\mathrm{SIM}}$ is the simulated MSE, $\widehat{\mathrm{V}}_{\mathrm{NBH}}$ is the local mean variance estimator and $\widehat{\mathrm{V}}_{\mathrm{LPM}}$ is LPM's variance estimator. Exact values are shown for SRS, $\mathrm{V}_{\mathrm{SRS}}$. The lowest simulated variance for each problem is shown in bold.

Pop	n	SRS	GRTS		GRTS ($2n$)		LPM			HIP		
		$\mathrm{V}_{\mathrm{SRS}}$	$\mathrm{V}_{\mathrm{SIM}}$	$\widehat{\mathrm{V}}_{\mathrm{NBH}}$	$\mathrm{V}_{\mathrm{SIM}}$	$\widehat{\mathrm{V}}_{\mathrm{NBH}}$	$\mathrm{V}_{\mathrm{SIM}}$	$\widehat{\mathrm{V}}_{\mathrm{NBH}}$	$\widehat{\mathrm{V}}_{\mathrm{LPM}}$	$\mathrm{V}_{\mathrm{SIM}}$	$\widehat{\mathrm{V}}_{\mathrm{NBH}}$	$\widehat{\mathrm{V}}_{\mathrm{LPM}}$
1	20	0.1044	0.0210	0.0282	0.0247	0.0279	0.0135	0.0306	0.0176	**0.0132**	0.0298	0.0160
	50	0.0416	0.0041	0.0053	0.0094	0.0051	0.0025	0.0057	0.0028	**0.0023**	0.0055	0.0026
	100	0.0207	0.0012	0.0014	0.0041	0.0014	0.0007	0.0015	0.0007	**0.0006**	0.0015	0.0007
	150	0.0137	0.0006	0.0006	0.0019	0.0006	0.0003	0.0007	0.0003	**0.0002**	0.0007	0.0003
	200	0.0103	0.0004	0.0004	0.0008	0.0003	**0.0002**	0.0004	0.0002	**0.0002**	0.0004	0.0002
2	20	0.1812	0.0801	0.1164	0.0875	0.1122	**0.0630**	0.1200	0.1219	0.0699	0.1162	0.1110
	50	0.0722	0.0171	0.0321	0.0285	0.0303	0.0143	0.0331	0.0238	**0.0131**	0.0331	0.0218
	100	0.0359	0.0071	0.0103	0.0121	0.0100	**0.0038**	0.0111	0.0063	0.0041	0.0108	0.0057
	150	0.0238	0.0036	0.0051	0.0069	0.0049	0.0021	0.0055	0.0029	**0.0019**	0.0053	0.0026
	200	0.0178	0.0018	0.0031	0.0036	0.0029	0.0012	0.0033	0.0016	**0.0010**	0.0031	0.0015
3	20	74.614	44.805	45.405	43.332	45.491	**30.875**	48.238	48.086	31.667	48.464	45.947
	50	29.756	9.016	13.304	12.275	12.913	**6.556**	13.760	10.557	7.937	13.590	9.843
	100	14.803	2.873	4.431	5.480	4.316	2.118	4.723	2.967	**1.686**	4.660	2.759
	150	9.819	1.448	2.215	3.096	2.099	1.031	2.376	1.389	**0.990**	2.316	1.262
	200	7.327	0.811	1.348	1.562	1.278	**0.602**	1.433	0.792	0.671	1.368	0.714

2 Populations used in Section 5

The sampling frame was a rectangular grid of $N = Z^2$ points in $[0, 1)^2$:

$$\left\{ \frac{1}{Z} \sum_{i=1}^{2} z_i \mathbf{e}_i : z_i \in \{0, 1, \ldots, Z-1\} \right\},$$

where $\mathbf{e}_i$ is the ith row of the two-dimensional identity matrix. Two Z values were considered: $Z = 30$ and $Z = 100$, giving point resources with 900 and 10,000 points, respectively. The response value for each point on the grid, $\mathbf{x} = (x_1, x_2)$, was defined as the integral of $f(\mathbf{x})$ over a box defined by

$$[x_1, x_1 + 1/Z) \times [x_2, x_2 + 1/Z).$$

Three functions were used to define response values, listed below.

- Population 1 (Robertson et al. 2013; Grafström, Lundström and Schelin 2012):

$$f(\mathbf{x}) = 3(x_1 + x_2) + \sin(6(x_1 + x_2)),$$

  with population total $\tau = 2.9994$ (see Figure 1(a)).

- Population 2 (Peak function):

$$\begin{aligned} f(\mathbf{x}) &= 3(4 - 6x_1)^2 \exp(-(6x_1 - 3)^2 - (6x_2 - 2)^2) \ldots \\ &-10(0.2(6x_1 - 3) - (6x_1 - 3)^3 - (6x_2 - 3)^5) \exp(-(6x_1 - 3)^2 - (6x_2 - 3)^2) \ldots \\ &-\frac{1}{3} \exp(-(6x_1 - 2)^2 - (6x_2 - 3)^2), \end{aligned}$$

  with population total $\tau = 0.3627$ (see Figure 1(b)).

- Population 3 (Bird function):

$$\begin{aligned} f(\mathbf{x}) = (12x_1 - 12x_2)^2 &+ \exp[(1 - \sin(12x_1 - 6))^2] \cos(12x_2 - 6) \ldots \\ &+ \exp[(1 - \cos(12x_2 - 6))^2] \sin(12x_1 - 6), \end{aligned}$$

  with population total $\tau = 23.3982$ (see Figure 1(c)).

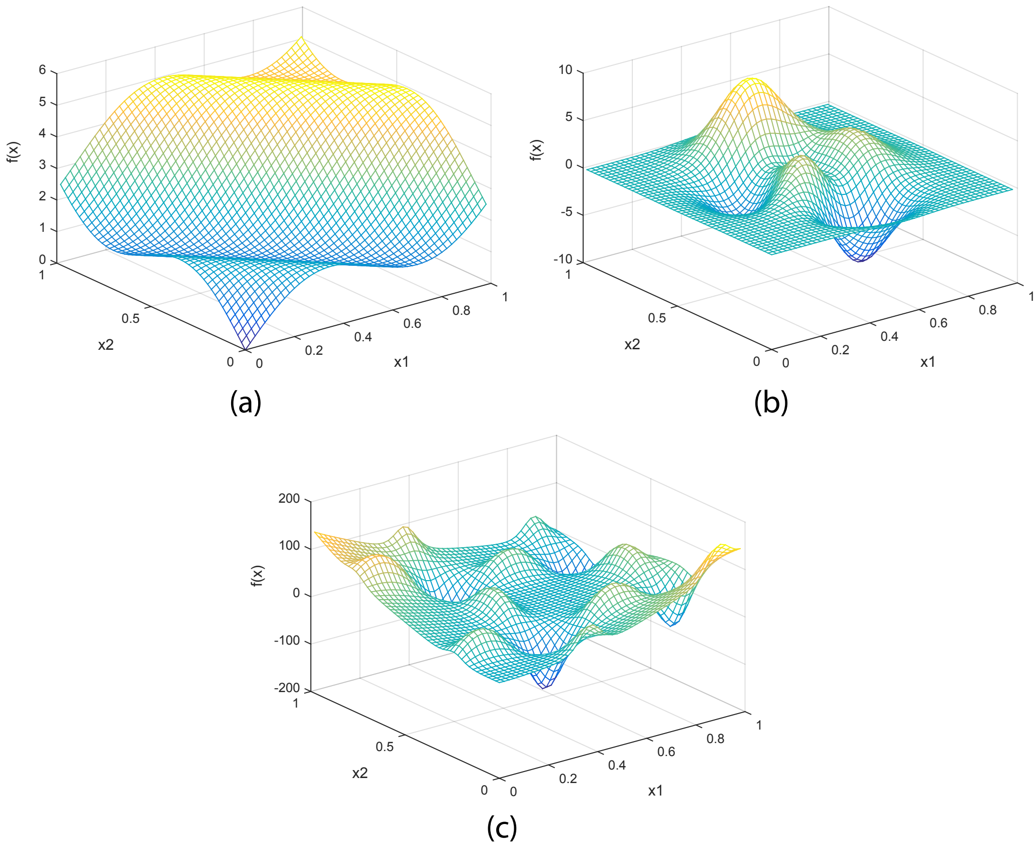

Figure 1: Functions used to define response values.

References

[1] Grafström, A., Lundström, N. L. P. and Schelin, L. (2012). Spatially balanced sampling through the pivotal method. *Biometrics* **68**, 514–520.

[2] Robertson, B. L., Brown, J. A., McDonald, T., and Jaksons, P. (2013). BAS: Balanced acceptance sampling of natural resources. *Biometrics* **3**, 776-784.